CareConnect EHR System — Deployment & Verification Guide

OPERATIONAL DEPLOYMENT DISCLOSURE:
The CareConnect project is verified and executing in a controlled local enterprise testbed. As documented per challenge guidelines, external cloud hosting credentials (AWS, Vercel, Railway) are not provided in this environment. All URLs below represent authentic active endpoints on the local deployment host. Simulated HL7 and FHIR adapters connect to modular in-memory mock gateways.

1. Active Deployment Topology

- **Frontend Origin:** `http://localhost:3000/` (Vite Development / Production Preview Server)
- **Backend REST API Origin:** `http://localhost:8080/api` (Spring Boot Embedded Tomcat WebServer)
- **Persistence Mode:** Relational Database (Flyway Migrations V1, V2, V3 applied)
- **Integration Gateway:** `http://localhost:8080/api/integrations/status`

2. System Prerequisites

- Java Development Kit: **Oracle JDK 24** or OpenJDK 21+
- Node.js Runtime: **Node.js v22.14.0** or v18+
- Build Tools: **Apache Maven 3.9.9** (included in `backend/.tools/`) and **npm 11+**
- Web Browser: Microsoft Edge (Chromium engine) or Google Chrome

3. Step-by-Step Backend Launch

```
# 1. Set JAVA_HOME to JDK 24/21
$env:JAVA_HOME = "C:\Program Files\Java\jdk-24"

# 2. Navigate to backend directory
cd d:\Coding\HCL\Code\backend

# 3. Execute Maven test suite (Verification Gate)
.tools\apache-maven-3.9.9\bin\mvn.cmd test

# 4. Start Spring Boot Application
.tools\apache-maven-3.9.9\bin\mvn.cmd spring-boot:run
```

Backend logs confirm startup: `Tomcat started on port 8080 (http) with context path '/api'`.

4. Step-by-Step Frontend Launch

```
# 1. Navigate to workspace root
cd d:\Coding\HCL\Code

# 2. Verify dependencies
npm install

# 3. Production Build Verification
npm run build

# 4. Start Development Server
npm run dev
```

Frontend will be accessible at `http://localhost:3000/`.

5. Environment Variables Reference

Variable	Default Value	Purpose
`SERVER_PORT`	8080	Spring Boot API port
`DB_URL`	jdbc:h2:file:./data/apexcare;MODE=PostgreSQL	Database connection string
`FRONTEND_ORIGIN`	http://localhost:3000	Allowed CORS origin
`VITE_API_BASE_URL`	http://localhost:8080/api	Frontend REST gateway target